

DEV ANILBHAI SHYARA

vasudevahir2006@gmail.com • linkedin.com/in/devshyara • github.com/Dev-6106 • Surat, India

Computer Science student at NIT Surat with expertise in data structures, algorithms, systems design, and full-stack development. Designed and deployed scalable software systems, including AI-powered knowledge platforms leveraging LLM orchestration, vector search, and distributed backend architectures. Seeking to apply strong problem-solving skills and systems-level thinking to build reliable, high-impact products at scale.

EDUCATION

B.Tech in Computer Science & Engineering

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat

GPA: 8.11 / 10.0

Expected May 2028

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, SQL

Backend: Node.js, Express.js, REST APIs, FastAPI

Frontend: React, Next.js, Redux, TailwindCSS, HTML/CSS

Databases: MongoDB, MySQL, Redis, Pinecone (Vector DB)

AI / ML: LangChain, LangGraph, Adaptive RAG, Cohere Embeddings, NumPy, Pandas, Matplotlib

Developer Tools: Git, GitHub, Docker, GitHub Actions, Vercel

Core CS: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, System Design

EXPERIENCE

AI / ML Engineering Intern — **Infosys Springboard** — Remote

2025

- Engineered a full-stack web application using FastAPI, Streamlit, and PostgreSQL to autonomously track, analyze, and visualize real-time e-commerce product prices.
- Developed an automated data extraction pipeline with BeautifulSoup and Selenium to continuously scrape, clean, and store product data from major online retailers into a relational database.
- Designed and evaluated an ensemble machine learning forecasting system using Random Forest and XGBoost to predict product price trends up to 30 days ahead, generating actionable *Buy*, *Wait*, and *Hold* recommendations.
- Integrated an LLM-powered conversational AI assistant and interactive analytics dashboards, enabling users to explore historical price volatility, optimize purchasing decisions, and configure personalized automated price alerts.

PROJECTS

NoteAtlas — AI Knowledge Operating System — <https://note-atlas.vercel.app/> · React · Express · MongoDB · Pinecone · LangChain · LangGraph · Cohere

- Built a full-stack AI knowledge platform (React 19 + Node.js/Express) ingesting PDFs, text, web links, and audio into a queryable knowledge graph persisted in MongoDB, with entity/relationship extraction via structured LLM output.
- Designed an **Adaptive RAG pipeline** using LangGraph—combining multi-query expansion, Pinecone vector retrieval with Cohere embeddings, Reciprocal Rank Fusion, Cohere Rerank scoring, and conditional web search fallback—for grounded Q&A over user documents.
- Built a **LangGraph-orchestrated Agent Studio** with 10 specialized AI workflows (research synthesis, fact verification, knowledge-gap analysis, timeline extraction, study planning, etc.) using map-reduce prompt chains with LRU-cached

LLM invocations.

- ▷ Added MongoDB compound indexes on common query patterns and implemented an in-process LRU cache for LLM responses to reduce redundant API calls.
- ▷ Deployed frontend on Vercel with a **GitHub Actions CI pipeline** (build verification for both frontend and backend on every push/PR to main); backend containerized via Docker.

RoadMate — Location-Based Roadside Assistance Platform

Next.js · TailwindCSS · MongoDB · Leaflet.js

- ▷ Designed and built a full-stack platform connecting drivers experiencing breakdowns with the nearest available mechanics, solving a real-time proximity-matching problem at the application layer using geospatial queries on MongoDB.
- ▷ Integrated **Leaflet.js** for an interactive map interface with live location visualization, enabling users to browse and request nearby mechanics with low-latency map tile rendering and marker updates.
- ▷ Engineered efficient service-discovery logic using MongoDB's `near` geospatial operator and indexed location fields, ensuring fast query responses as dataset size scales.
- ▷ Built a fully responsive UI with Next.js and TailwindCSS, achieving consistent UX across mobile and desktop form factors; optimized Core Web Vitals with server-side rendering and image optimization.
- ▷ Implemented RESTful API routes in Next.js API handlers with input validation and structured error responses, following production API design conventions.